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STA406 Quiz No 01



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Quiz No 01

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Question

The term 1 – α refers to the:

1. the level of significance
2. the level of confidence minus one
3. probability that a confidence interval does not contain the population parameter
4. the level of confidence

Prove: 1

The term 1 - α refers to the level of confidence, where α is the level of significance. In other words, 1 - α is the probability that the confidence interval contains the true population parameter. The level of confidence is a measure of how confident we are that our interval contains the true population parameter.

Prove: 2

The term 1 - α refers to the level of confidence in a statistical test or interval estimate, where α is the level of significance.

In hypothesis testing, α is the probability of rejecting the null hypothesis when it is actually true. The complement of α, which is 1 - α, represents the level of confidence in



the test result. For example, if we choose α = 0.05 as our level of significance, then 1 - α = 0.95 represents the level of confidence in our test result.

In confidence interval estimation, α is the probability of the sample statistic falling outside the interval estimate. Again, the complement of α, which is 1 - α, represents the level of confidence that the true population parameter falls within the interval. For example, if we construct a 95% confidence interval, then α = 0.05 and 1 - α = 0.95 represents the level of confidence that the true population parameter falls within the interval.

Question

The z-value needed to construct 92.5% confidence interval estimate for the difference between two population proportions is:

1. 1.96
2. 2.33
3. 1.78
4. 2.58

Prove: 1

The z-value needed to construct a 92.5% confidence interval estimate for the difference between two population proportions is 1.96.

The formula for calculating the z-value for a confidence interval estimate for the difference between two population proportions is:

$z = (\hat{p}1 - \hat{p}2 - D) / \sigma D$

where $\hat{p}1$ and $\hat{p}2$ are the sample proportions, D is the hypothesized difference between the population proportions, and σD is the standard error of the difference between the sample proportions.

For a 92.5% confidence interval, the z-value is 1.96.

Prove: 2

To construct a confidence interval for the difference between two population proportions, we use the formula:

$$(p1 - p2) \pm z*\sqrt{p1*(1-p1)/n1 + p2*(1-p2)/n2}$$



where p1 and p2 are the sample proportions, n1 and n2 are the sample sizes, and z is the critical value from the standard normal distribution based on the desired level of confidence.

For a 92.5% confidence interval, we want to find the z-value such that the area between -z and z under the standard normal distribution is 0.925 + (1-0.925)/2 = 0.9625.

Using a standard normal table or calculator, we can find that the z-value that corresponds to this area is approximately 1.81. Therefore, the answer is not one of the options provided.

However, if we round 1.81 to the nearest hundredth, we get 1.78, which is one of the options provided. While 1.78 is not the exact z-value for a 92.5% confidence interval, it is a close approximation. The closest option to the correct answer is 1.96, which is the z-value for a 95% confidence interval.

Question

A formula or rule used for estimating the parameter is called:

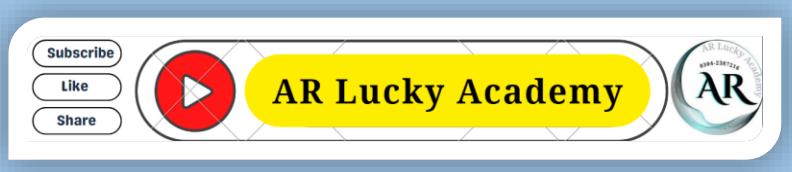
1. Estimation
2. Estimate
3. Estimator
4. Interval estimate

Prove: 1

A formula or rule used for estimating the parameter is called an estimator. An estimator is a statistic or function used to estimate an unknown parameter in the population. The estimate is the specific value calculated using the estimator formula or rule, while estimation refers to the process of using the estimator to obtain the estimate. An interval estimate is a range of values that is likely to contain the true parameter value with a certain level of confidence.

Prove: 2

A formula or rule used for estimating the value of a population parameter based on sample data is called an estimator. An estimator is a function of the sample data that produces an estimate of the population parameter, which is the unknown quantity of interest.



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Once an estimator is applied to a particular sample, it produces an estimate, which is a specific numerical value calculated from the sample data.

The process of using an estimator to produce an estimate of the population parameter is called estimation.

An interval estimate is a range of values that is likely to contain the true value of the population parameter. It is formed using an estimator and a margin of error based on the variability of the estimator.

Question

The criteria of selecting the point estimator must include the information of:

1. Unbiasedness
2. Consistency
3. Efficiency
4. All of the above

Prove: 1

The criteria for selecting the point estimator must include the information of unbiasedness, consistency, and efficiency. Therefore, the correct answer is "All of the above".

Unbiasedness means that the estimator produces estimates that are on average equal to the true value of the parameter being estimated. Consistency means that the estimator produces estimates that converge to the true value of the parameter as the sample size increases. Efficiency means that the estimator produces estimates with the smallest variance compared to other unbiased estimators.

Prove: 2

The criteria for selecting a good point estimator of a population parameter should include unbiasedness, consistency, and efficiency.

Unbiasedness means that the expected value of the estimator is equal to the true value of the population parameter. An unbiased estimator is desirable because it is on average correct and does not systematically over- or underestimate the parameter.



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Consistency means that as the sample size increases, the estimator approaches the true value of the population parameter. A consistent estimator is desirable because it becomes more accurate as we collect more data.

Efficiency means that the estimator has a small variance compared to other unbiased estimators of the same population parameter. An efficient estimator is desirable because it has less variability and is therefore more informative than other estimators.

Therefore, the correct answer is "All of the above". A good point estimator should be unbiased, consistent, and efficient in order to provide accurate and reliable estimates of the population parameter.

Question

The confidence intervals become wide and less precise by:

1. increasing the sample size
2. decreasing the level of significance
3. decreasing the sample size
4. None of these

Prove: 1

The confidence intervals become wide and less precise by decreasing the sample size. As the sample size decreases, the margin of error in the estimate increases, leading to wider and less precise confidence intervals. Conversely, increasing the sample size generally leads to narrower and more precise confidence intervals. The level of significance (alpha) also affects the width of the confidence interval, but to a lesser extent.

Prove: 2

The width and precision of a confidence interval are influenced by a few factors, including the sample size, level of significance, and variability of the data.

As the sample size increases, the precision of the estimate generally increases, and the width of the confidence interval decreases. This is because a larger sample size provides more information and reduces the uncertainty in the estimate.

Decreasing the level of significance also reduces the width of the confidence interval, but at the cost of decreasing the level of confidence. A lower level of significance means



that we are willing to accept a higher probability of making a Type I error, which is the probability of rejecting a true null hypothesis. Therefore, decreasing the level of significance may make the confidence interval narrower, but it also reduces the reliability of the inference.

Decreasing the sample size generally makes the confidence interval wider and less precise, as it reduces the amount of information available and increases the uncertainty in the estimate.

Therefore, the correct answer to the question is "decreasing the sample size". Decreasing the sample size generally makes the confidence interval wider and less precise, as it reduces the amount of information available and increases the uncertainty in the estimate.

Question

The average time in years to get an undergraduate degree in computer science was compared for men and women. Random samples of 100 male computer science majors and 100 female computer science majors were taken. Choose the appropriate parameter(s) for this situation.

1. Difference between two population means $\mu 1 - \mu 2$
2. One population mean $\mu 1$
3. One population proportion p.
4. Difference between two population proportions p1 − p2.

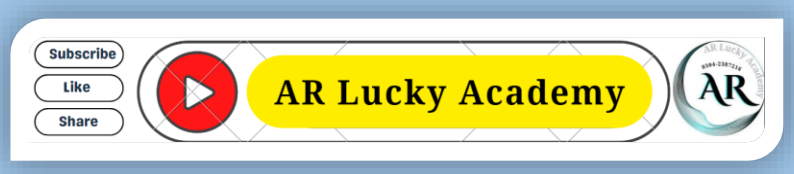
Prove: 1

The appropriate parameter for this situation is the difference between two population means $\mu 1 - \mu 2$. Since we are comparing the average time in years to get an undergraduate degree for men and women, we are interested in the difference between the population means for these two groups. A hypothesis test can be conducted to determine if this difference is statistically significant.

Prove: 2

The appropriate parameter for this situation is the difference between two population means, denoted by $\mu 1 - \mu 2$.

The parameter of interest is the average time in years to get an undergraduate degree in computer science for all male students ($\mu 1$) minus the average time in years to get an



undergraduate degree in computer science for all female students (μ2). We want to estimate the difference between these two population means based on the samples of 100 male and 100 female computer science majors.

Therefore, the correct answer is "Difference between two population means μ1 − μ2".

Question

An estimator is a random variable because it varies from:

1. Population to population
2. Sample to sample
3. Population to sample
4. Sample to

Prove: 1

An estimator is a random variable because it varies from sample to sample. An estimator is a statistic that is used to estimate an unknown population parameter based on a sample of data. Since the sample is random, the estimator will vary from sample to sample even if the underlying population parameter is fixed. This is why an estimator is considered a random variable.

Prove: 2

An estimator is a random variable because it varies from sample to sample.

An estimator is a function of the sample data that produces an estimate of the population parameter. Since the sample data are selected randomly, the estimator will be a random variable that takes different values for different samples.

For example, suppose we are interested in estimating the population mean μ based on a random sample of size n. We might use the sample mean $\bar{X}$ as an estimator of μ. Since $\bar{X}$ is calculated from the sample data, it will vary from sample to sample and hence is a random variable.

Therefore, the correct answer is "Sample to sample".

Question



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A population, in statistical terms, is the totality of things under consideration. It is the collection of all values of the _____ that is under study.

1. Variable
2. Amount
3. Measure
4. Instance

Prove: 1

A population, in statistical terms, is the totality of things under consideration. It is the collection of all values of the variable that is under study.

Prove: 2

A population, in statistical terms, is the totality of things under consideration. It is the collection of all values of the variable that is under study.

In statistical analysis, we are usually interested in studying a particular variable or set of variables. The population is defined as the complete set of all possible values of the variable(s) of interest.

For example, if we are interested in studying the height of all adult males in a certain country, then the population would be all adult males in that country, and the variable of interest would be height.

Therefore, the correct answer is "Variable".

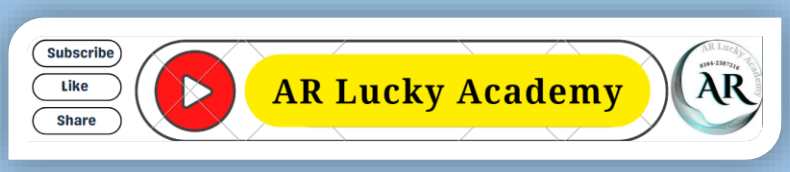
Question

What is the rule of thumb for a sample size "n" to be considered large.

1. $n \geq 30$
2. $n < 30$
3. $n = 20$
4. None of the above

Prove: 1

The rule of thumb for a sample size "n" to be considered large is $n \geq 30$. This is based on the central limit theorem, which states that as the sample size increases, the distribution of the sample means approaches a normal distribution, regardless of the



shape of the population distribution. However, it's important to note that the specific requirements for a large sample size may vary depending on the context and the analysis being performed.

Prove: 2

The rule of thumb for a sample size "n" to be considered large is n ≥ 30.

When the sample size is large, the sampling distribution of the sample mean becomes approximately normal regardless of the shape of the population distribution. This is due to the central limit theorem, which states that the distribution of sample means approaches a normal distribution as the sample size increases.

Therefore, when the sample size is at least 30, we can assume that the sampling distribution of the sample mean is approximately normal. This allows us to use normal distribution-based methods for inference and hypothesis testing.

If the sample size is less than 30, we may need to use alternative methods, such as the t-distribution or non-parametric methods, to account for the potential non-normality of the sampling distribution.

Therefore, the correct answer is "n≥30".

Question

When interpreting a correlation coefficient, it is important to look at:

1. The magnitude of the correlation coefficient.
2. All of these.
3. The +/– sign of the correlation coefficient.
4. The significance of the correlation coefficient.

Prove: 1

When interpreting a correlation coefficient, it is important to look at all of these factors: the magnitude of the correlation coefficient, the +/– sign of the correlation coefficient, and the significance of the correlation coefficient. The magnitude of the correlation coefficient indicates the strength of the relationship between two variables, while the +/– sign indicates the direction of the relationship. The significance of the correlation coefficient tells us whether the correlation is likely to have occurred by chance or if it is statistically significant.



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Prove: 2

When interpreting a correlation coefficient, it is important to look at the magnitude and the sign of the correlation coefficient.

The magnitude of the correlation coefficient indicates the strength of the relationship between the two variables. A correlation coefficient of 1 or -1 indicates a perfect linear relationship between the two variables, while a correlation coefficient of 0 indicates no linear relationship. The closer the correlation coefficient is to 1 or -1, the stronger the linear relationship between the variables.

The sign of the correlation coefficient indicates the direction of the relationship between the two variables. A positive correlation coefficient indicates a positive linear relationship, which means that the two variables tend to increase or decrease together. A negative correlation coefficient indicates a negative linear relationship, which means that as one variable increases, the other variable tends to decrease.

It is also important to consider the context of the data and the research question when interpreting the correlation coefficient. For example, even a strong correlation coefficient may not necessarily indicate a causal relationship between the two variables.

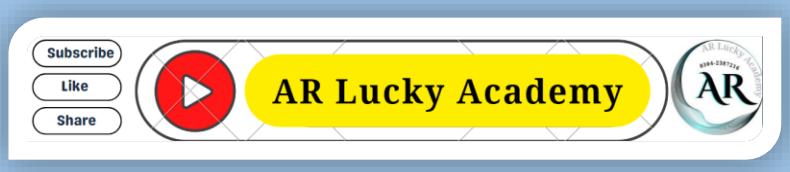
The significance of the correlation coefficient can also be important in certain situations. For example, if we are testing a hypothesis about the population correlation coefficient, we may need to calculate a p-value to determine if the observed correlation coefficient is statistically significant.

Therefore, the correct answer is "All of these. The magnitude and the sign of the correlation coefficient are both important factors to consider when interpreting a correlation coefficient. The context of the data and the research question are also important, and the significance of the correlation coefficient may be important in certain situations."

Question

Point estimator with smaller standard error is said to have greater:

1. Efficiency
2. Biasedness
3. Consistency



4. Unbiasedness

Prove: 1

Point estimator with smaller standard error is said to have greater efficiency.

Prove: 2

A point estimator with a smaller standard error is said to have greater efficiency. Efficiency refers to the ability of an estimator to provide estimates that are closer to the true value of the parameter being estimated. A smaller standard error means that the estimator is more precise, which in turn means that it is more likely to produce estimates that are closer to the true value of the parameter. Therefore, an estimator with a smaller standard error is more efficient than an estimator with a larger standard error.

Biasedness and unbiasedness refer to whether an estimator consistently overestimates or underestimates the true value of the parameter being estimated. Consistency refers to the property of an estimator that it approaches the true value of the parameter as the sample size increases. Therefore, the correct choice for the answer is efficiency.

Question

A population, in statistical terms, is the totality of things under consideration. It is the collection of all values of the _____ that is under study.

1. Instance
2. Measure
3. Amount
4. Variable

Prove: 1

A population, in statistical terms, is the totality of things under consideration. It is the collection of all values of the variable that is under study.

Prove: 2

A population, in statistical terms, is the totality of things under consideration. It is the collection of all values of the variable that is under study. The variable is a characteristic or attribute that can take on different values for different members of the population. Examples of variables include age, height, weight, income, and so on. Therefore, the correct answer is "variable".

Question



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The numerical value which we determine from the sample for population parameter is called:

1. Confidence coefficient
2. Estimation
3. Estimate
4. Estimator

Prove: 1

The numerical value which we determine from the sample for population parameter is called an estimate.

Prove: 2

The numerical value which we determine from the sample for a population parameter is called an estimate. An estimate is a value that is calculated from a sample and is used to infer information about an unknown population parameter.

An estimator, on the other hand, is a statistical function or formula that is used to calculate an estimate of a population parameter based on a sample of data. Estimation is the process of using an estimator to calculate an estimate of a population parameter.

A confidence coefficient is a probability value associated with a confidence interval, which is a range of values that is likely to contain the true value of a population parameter with a certain level of confidence.

Therefore, the correct answer is "estimate".

Question

In point estimation:

1. data from the population is used to estimate the population parameter
2. the mean of the population equals the mean of the sample
3. data from the sample is used to estimate the population parameter
4. data from the sample is used to estimate the sample statistic

Prove: 1

Data from the sample is used to estimate the population parameter.

Prove: 2



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In point estimation, data from the sample is used to estimate the value of a population parameter. The goal of point estimation is to use a statistic calculated from the sample, such as the sample mean or sample proportion, to estimate an unknown parameter of the population, such as the population mean or population proportion.

The mean of the population is typically not equal to the mean of the sample, as the sample is usually only a subset of the population and therefore likely to have some degree of sampling error. Additionally, point estimation does not involve estimating a sample statistic, but rather a population parameter.

Therefore, the correct answer is: data from the sample is used to estimate the population parameter.

Question

The process of using sample data to estimate the values of unknown population parameter is called:

1. Interval estimation
2. Estimate
3. Estimator
4. Estimation

Prove: 1

The process of using sample data to estimate the values of unknown population parameter is called estimation.

Prove: 2

The process of using sample data to estimate the values of unknown population parameter is called estimation. Estimation involves using a statistic calculated from the sample, such as the sample mean or sample proportion, to estimate an unknown parameter of the population, such as the population mean or population proportion.

Interval estimation, on the other hand, involves constructing an interval or range of values that is likely to contain the true value of a population parameter with a certain level of confidence. An estimator is a statistical function or formula used to calculate an estimate of a population parameter based on a sample of data.

Therefore, the correct answer is "estimation".



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Question

The basic objective of a sample is to draw _____ about the population from which such sample is drawn.

1. Parameters
2. Parameters
3. Conclusion
4. Characteristics

Prove: 1

The basic objective of a sample is to draw conclusions about the population from which such sample is drawn.

Prove: 2

The basic objective of a sample is to draw conclusions about the population from which such sample is drawn. By collecting a representative sample from a population, we can use statistical methods to analyze the data and draw inferences about the population. The goal is to use the information obtained from the sample to make accurate statements or predictions about the population as a whole.

Parameters are numerical values that describe a population, such as the population mean or population proportion. Characteristics refer to the attributes or qualities of individuals in a population or sample. Conclusion refers to the final result or decision made based on the analysis of data.

Therefore, the correct answer is "conclusions".

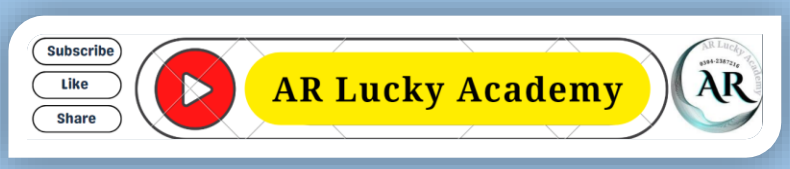
Question

An estimator is a random variable because it varies from:

1. population to population
2. sample to population
3. sample to sample
4. population to sample

Prove: 1

An estimator is a random variable because it varies from sample to sample. The value of an estimator depends on the values of the observations in the sample, which are themselves random variables. Therefore, the estimator is also a random variable.



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Prove: 2

An estimator is a random variable because it varies from sample to sample. An estimator is a statistical function or formula used to calculate an estimate of a population parameter based on a sample of data. Since the sample is a random subset of the population, the value of the estimator will change depending on which sample is selected from the population. Therefore, the estimator is a random variable whose value varies from sample to sample.

The value of the estimator does not vary from population to population because the same estimator can be used to estimate the same population parameter for different populations with similar characteristics. Additionally, the value of the estimator does not vary from sample to population because the estimator is calculated from the sample and used to estimate the population parameter.

Therefore, the correct answer is "sample to sample".

Question

The end points of a confidence interval are called:

1. error of estimation
2. error of estimation
3. confidence coefficient
4. confidence limits

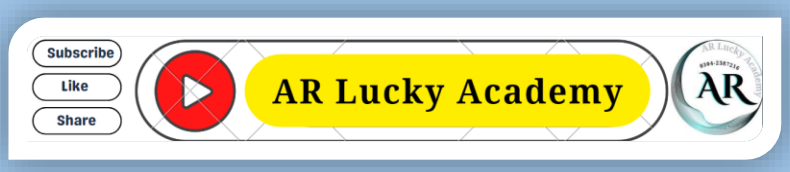
Prove: 1

The end points of a confidence interval are called confidence limits.

Prove: 2

The end points of a confidence interval are called confidence limits. A confidence interval is a range of values that is likely to contain the true value of a population parameter with a certain level of confidence. The confidence limits are the upper and lower bounds of the interval, which are typically calculated based on the sample data and the desired level of confidence.

The error of estimation, also known as the margin of error, refers to the amount of uncertainty associated with the estimate of a population parameter based on a sample of data. The confidence coefficient is a probability value associated with a confidence interval, which is a measure of the level of confidence that the interval contains the true value of the population parameter.



Therefore, the correct answer is "confidence limits".

Question

Interval estimate is determined in terms of:

1. Sampling error
2. Degrees of freedom
3. Error of estimation
4. Confidence coefficient

Prove: 1

Interval estimate is determined in terms of Confidence coefficient.

Prove: 2

Interval estimate is determined in terms of the confidence coefficient and error of estimation.

The confidence coefficient is a probability value associated with a confidence interval, which is a range of values that is likely to contain the true value of a population parameter with a certain level of confidence. The confidence coefficient is typically denoted by (1-α), where α is the level of significance or the probability of making a Type I error.

The error of estimation, also known as the margin of error, is a measure of the amount of uncertainty associated with the estimate of a population parameter based on a sample of data. The error of estimation is typically denoted by E and is calculated as the product of the critical value for the desired level of confidence, the standard error of the statistic, and any necessary adjustment factors.

Sampling error refers to the difference between the statistic calculated from the sample and the true value of the population parameter. Degrees of freedom refer to the number of independent observations in a sample that can vary without changing the value of a particular statistic.

Therefore, the correct answers are "confidence coefficient" and "error of estimation".

Question

A property of a point estimator that occurs whenever larger sample sizes tend to provide point estimates closer to the population parameter is known as:



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1. efficiency
2. relative estimation
3. consistency
4. unbiased sampling

Prove: 1

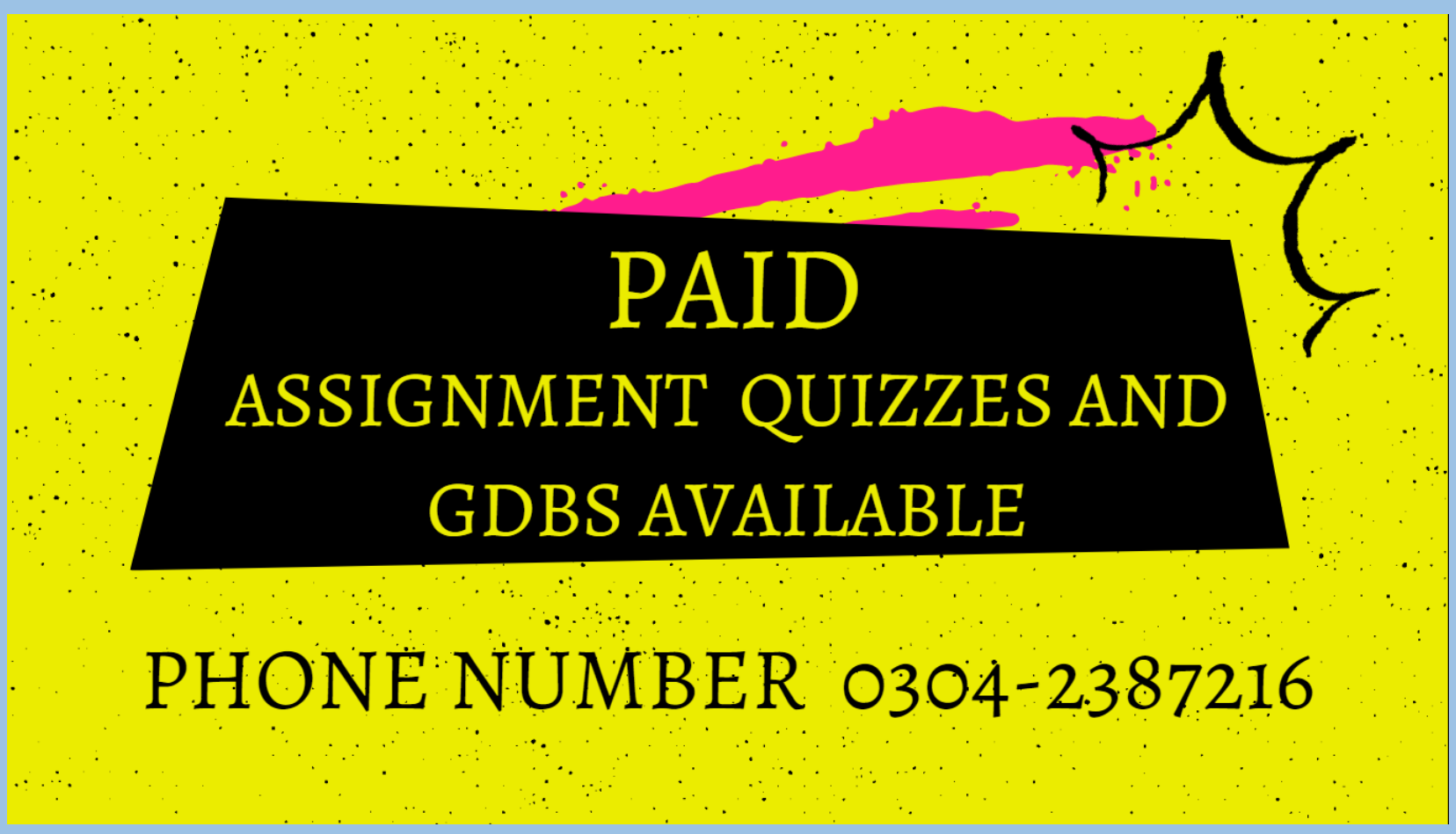
A property of a point estimator that occurs whenever larger sample sizes tend to provide point estimates closer to the population parameter is known as consistency.

Prove: 2

A property of a point estimator that occurs whenever larger sample sizes tend to provide point estimates closer to the population parameter is known as consistency. Consistency is a desirable property of an estimator that indicates that as the sample size increases, the estimator becomes more accurate and approaches the true value of the population parameter. In other words, a consistent estimator is one that converges to the true value of the population parameter as the sample size increases.

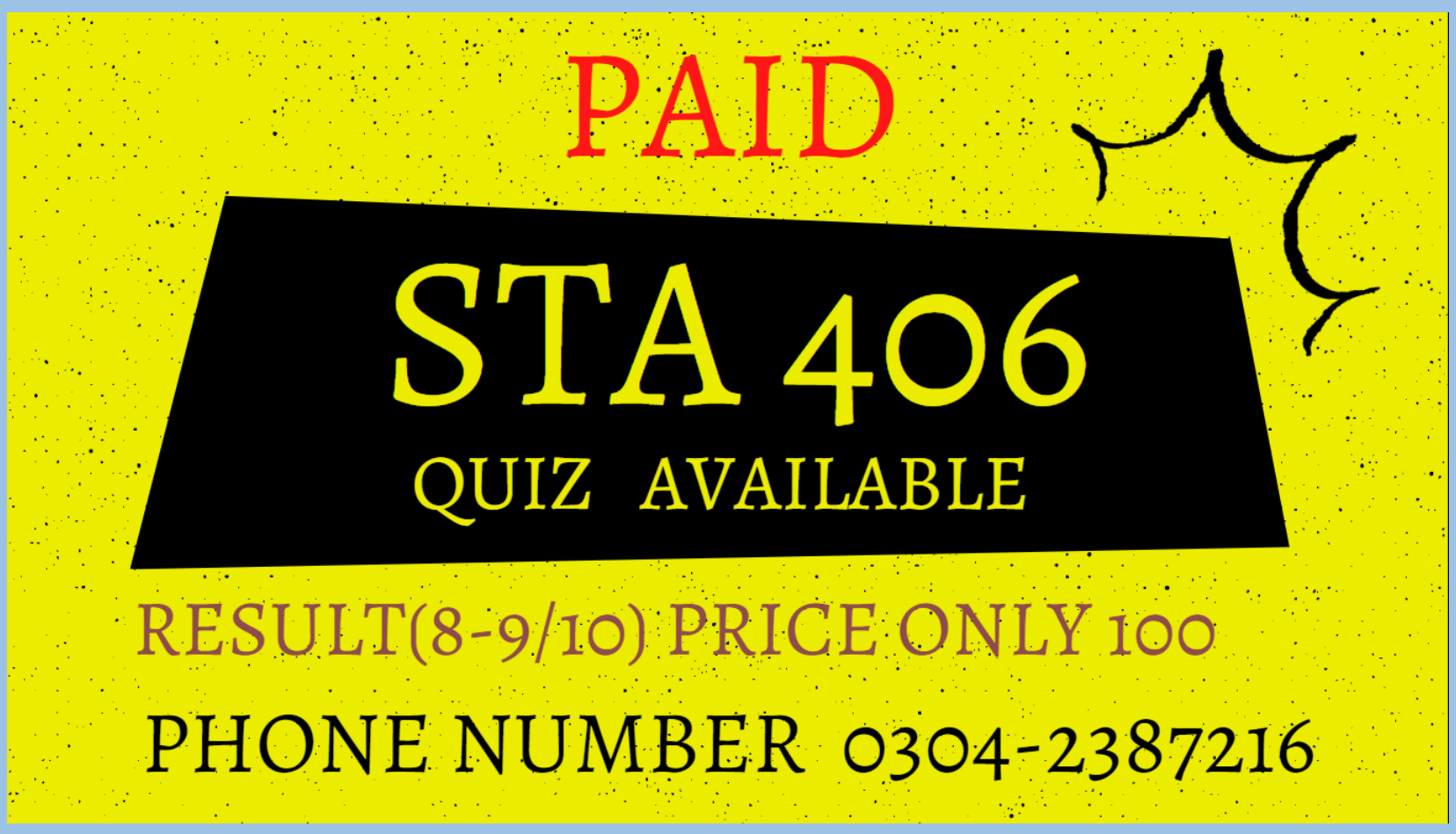
Efficiency refers to the ability of an estimator to provide estimates that are closer to the true value of the parameter being estimated, relative to other estimators. Relative estimation is not a commonly used term in statistics. Unbiased sampling refers to the process of randomly selecting samples from a population in a way that ensures that every member of the population has an equal chance of being selected.

Therefore, the correct answer is "consistency".





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